Allsky GO



User Manual Gen2.

October 2026



Allsky GO

Contents

3D Printed Design & Layout 3

Introducing the Allsky *GO!* Timelapse Camera 4

Pursuit of Dark Skies 5

Scope of Development 6

Key Hardware Components 9

Autonomy 11

System Set-up 12

Software & Scripts 13

Bill of Material (3D Printed) 14

Dome Assembly (V3 for PiHQ or ZWO) 14

AllskyGo Tub Assy 15

AllskyGo Smart Base & Cradle 15

Bill of Material (Non-Printed) 16

Other Resources & Links 17



Allsky GO

3D Printed Design & Layout



Allsky GO

Introducing the Allsky *GO!* Timelapse Camera



Inspired by the Thomas Jacquin concept, Allsky Go is the latest innovation in all-sky camera technology; designed to be printed from home for seamless use both on and off-grid. This versatile camera is built to capture the night sky wherever your curiosity takes you. Originally launched in 2025 – the 2nd Generation design offers additional flexibility and performance improvements



1st generation (2025) AllskyGo



Allsky GO

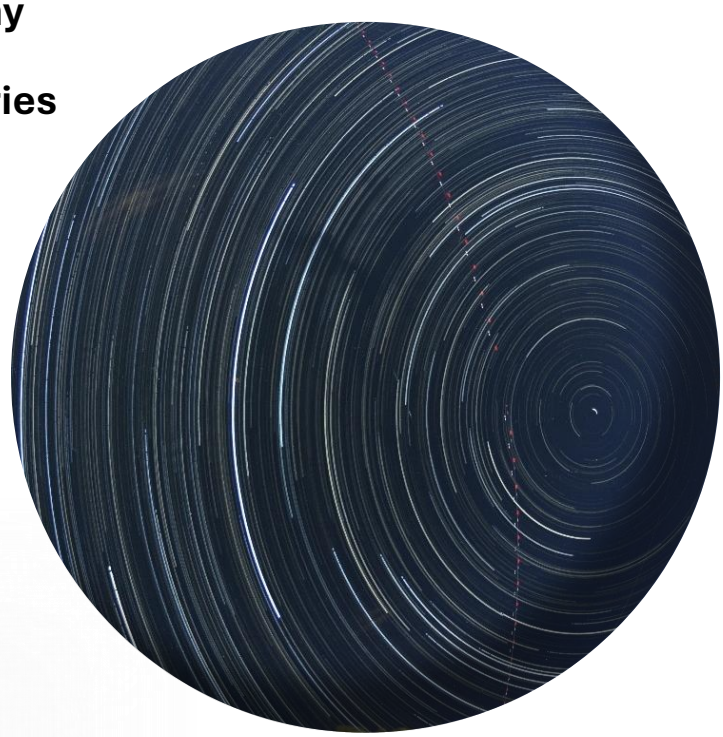
🌌 Allsky Go!

Pursuit of Dark Skies

Whether you are camping under remote starlit skies, exploring new destinations, or gazing from your own backyard, Allsky Go! is your cosmic companion. Ready to reveal the magic of the night wherever you roam. Discover the stars like never before with a compact, smart, and rugged system built for travel and exploration.

Your Portal to the Night Sky

- 🏕️ **Camping Expeditions**
- 🌩️ **Storm Chasing Adventures**
- ✈️ **Plane Spotting Missions**
- 🌠 **Dark-Sky Photography**
- 🏡 **Backyard Observatories**
- 🏫 **Education**





Allsky GO

Scope of Development

The core mission behind Allsky Go!

- 📷 **Core System**
  - Standard Allsky setup running Pi4 Bookworm
  - Recommend TJ Open-source software
  - PI HQ & ZWO Compatible V3 Allsky Dome
- ✈ **Mobility & Independence**
  - Fully portable design with >12-hour autonomy
  - Travel Safe Battery, rechargeable via USB.
  - 15W USB Power Input, Off-grid capable
- 🌀 **Environmental Awareness**
  - RTC off-grid time synchronization
  - GPS for geolocation
  - Humidity and temperature sensors
- 🔌 **Power & Connectivity**
  - USB-C charging port
  - WLAN & LAN support for connectivity on the fly



Allsky GO

Allsky GO → NEW Allsky GO



New Styling

- Latest V3 Dome Assembly
- Compatible with Multi-Material Printers





NEW

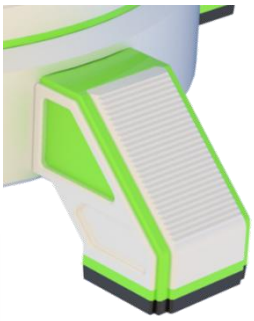
Improved 2nd Gen Smart Base

- Reduced Overhangs / Few Supports
- Ethernet/POE Port, Customizable Function & Power Buttons
- Tripod Thread Insert, External 3.5mm Jack Port (Optional)

NEW

Additional Features

- Improved Stability Landing Feet
- Integrated Humidity Probe





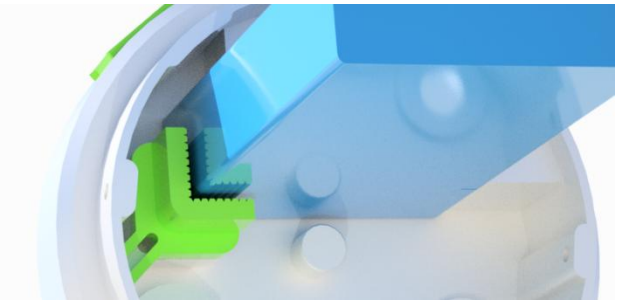


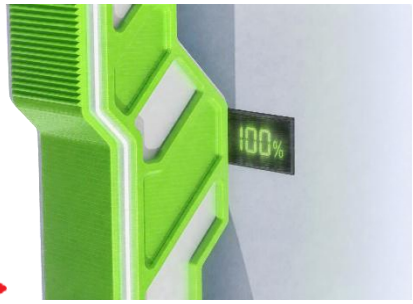
Allsky GO

NEW

Powerbank

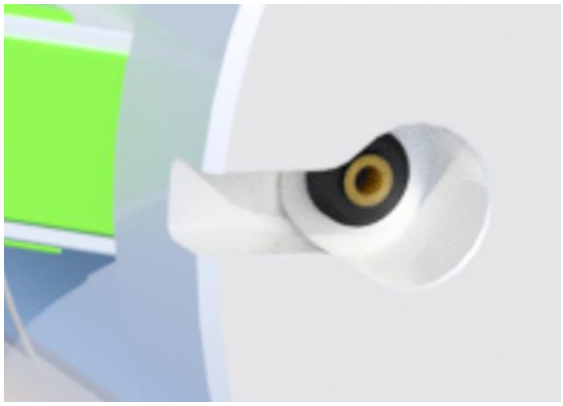
- Adjustable clamp to support locally available Powerbank
- 15W Supply optimized for Raspberry Pi4b





NEW

Clear PETG window to inspect charge level (experimental)



NEW

Sensor Expansion

- Optional 4-Pole Jack port for sensor expansion



NEW

GPS Antenna

- Integrated GPS Antenna Position

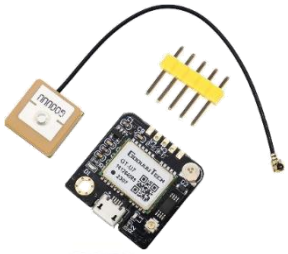
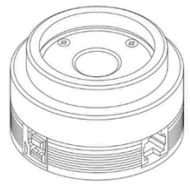
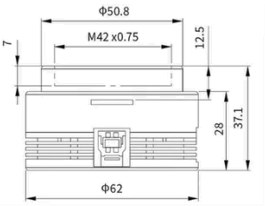


Allsky GO

Key Hardware Components







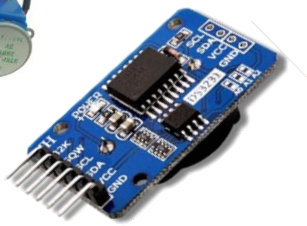










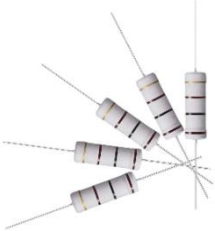














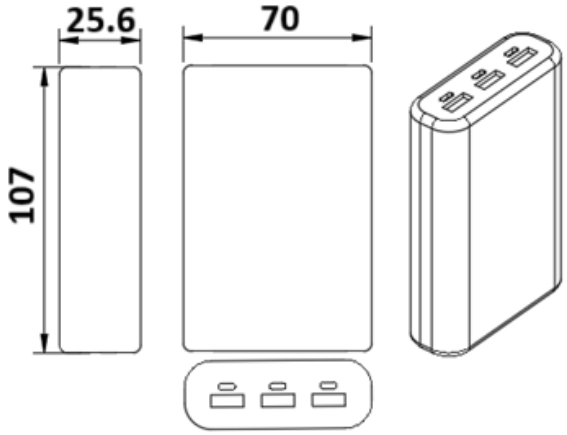

Allsky GO

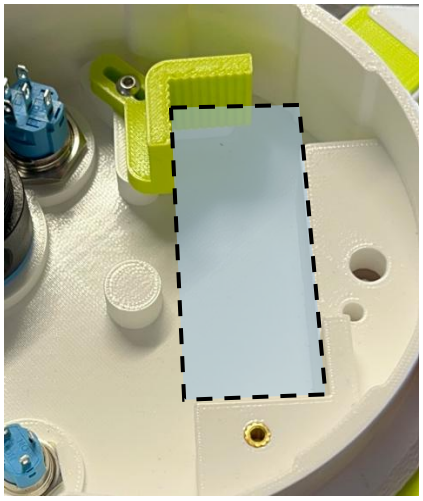
Power bank



Designed around the VRURC 20.000mah Power bank for compact size and versatility.

Available to purchase in EU and NA for around $30 (Amazon.it | Amazon.co.uk | Amazon.com)





Allskygo Clamping mechanism designed to accommodate **other powerbank** solutions with the following key dimensions;

Powerbank Dimensions	**Length**	**Width**	**Height**
Mix	65.5mm	22.5mm	≈130mm
Max	71.5mm	27.5mm	≈130mm



Allsky GO

Powerbank must supply 15W @ 5V 3amp to satisfy Raspberry Pi4 power requirement, and support charging and discharging on the same USB-C Port (In/Out)

*Raspberry Pi5 not recommended due to 25W Supply Demand (beyond current USB standard protocol)

Autonomy

Measured Autonomy exceeds **16hrs continuous** use on a single charge. Battery recharges within 8hrs using fast charging port.

*(*Dew Heating, or ZWO Camera my effect total Autonomy)*.



Allsky GO

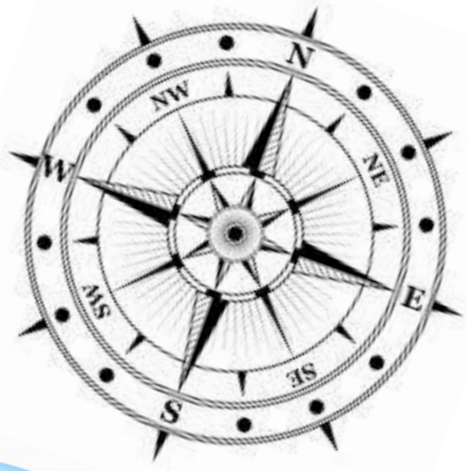
System Set-up

The goal of Allsky Go! a one-button setup, just switch it on and let it do the rest.

Thanks to its **integrated RTC Battery** & **GPS receiver (GT-U7)**, it can determine your time and location, updating the coordinates overlay automatically without user input; GPS coordinates are also used to adjust the switch between day and night in the Allsky settings.json

The **temperature and humidity sensor (SHT31**) can also be programmed to activate secondary circuits or heating elements to add further protection against frost or dew.







Allsky GO

Software & Scripts

The system runs on standard Allsky Software available from the Allsky Team Repository.

I share custom scripts for initial button and sensor integration through a specific Allsky Go GitHub site.

Allsky_Go/AllskyGo Gen2 at main · Chrise-2000/Allsky_Go

- **Button Service**

  Background service monitoring button press for Short & Long Press Button Function:

  - *Button A: WPS One-Press guest connection*
  - *Button A: Safe Shutdown (Hold > 2sec)*
  - *Button B: GenerateForDay (Allsky End of Night Sequence)*
- **Buzzer Service - Start-up Melodies**

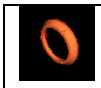
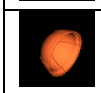


Allsky GO

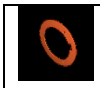
Bill of Material (3D Printed)



Dome Assembly (V3 for PiHQ or ZWO)

	Dome Ring	Allsky V3.1 - Dome Ring Rain Improvement
	Camera Cone	Allsky V3.0 - PiHQ Camera Cone Vented v1.0
	Camera Cone	Allsky V3.0 - ZWO Camera Cone Vented v1.0
	Ring Vents	Allsky V3.0 - Ring Vents v1.0
	Ventilation Body	Allsky V3.0 - Ventilation Body Standard v1.0
	Zwo Retainer	Allsky V2.0 - ZWO Retainer v2 - 30mm Fan
	V3 Dome Cover	Allsky Dome Protection for V3

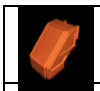
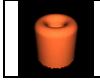
Seals and Gaskets (TPU)

	Seals	Allsky-v1.1-gasket-top-tpu
	Seals	Ethernet Cap Seal v1
	Seals	AllskyGo Gen2 MainSeal

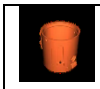


Allsky GO

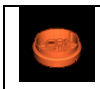
Feet

	Base	AllskyGo Foot Assy v1.0
	Base	AllskyGo Foot Plug v1.0
	Base	Metal M3 Stud (Optional)

AllskyGo Tub Assy

	Tub	AllskyGo Tub Assy v1.0
	Tub	AllskyGo Dew Sensor Probe v1.0
	Tub	AllskyGo Dew Sensor Clip

AllskyGo Smart Base & Cradle

	Base	AllskyGo SmartBase v1.1
	Cradle	AllskyGo LowerCradle v1.1
	Cradle	AllskyGo UpperCradle v1.1
	Base	AllskyGo Powerbank Clamp v1.0
	Base	Ethernet Cap v1



Allsky GO

Bill of Material (Non-Printed)

V3 Dome, Raspberry & Camera

- JMX 3.5 Inch Dome - Amazon
- Pi HQ Camera Module - Amazon / Reichelt / ZWO ASI Sensor
- Pi4 Camera Cable – 30cm (15pin) - Amazon / 90 USB3 TypeB Down Connector
- CS Lens 2.5 - 2.8mm / 150-180 degree FoV - AliExpress

Required Electronics

- Raspberry Pi4B 4GB Memory, MicroSD Card - Amazon
- Power Bank 20.000mah - Amazon (VRURC Mini Power bank)
- 30x30 5v Fan - Amazon
- Momentary Push Button (M12 Thread, Stainless, 3v-5v) x 2 - Amazon
- Power Button (M16 Thread, Stainless, 3v-5v) - Amazon
- USBC-USBC Passthrough - Amazon
- RJ45-RJ45 Passthrough - Amazon
- USBC-USBC Cable with 90degree connector - Amazon

Optional Electronics

- Recommended: GPS Module (GT-U7) - Amazon
- Recommended: Piezoelectric BIOS Speaker - Amazon
- Recommended: RTC Real Time Clock DS3231 - Amazon
- Recommended: Humidity Sensor (SHT31) (Optional) - Amazon
- Option: 100ohm Resistors Metal Oxide Film (Optional) - Amazon
- Option: Stepper Motor & Driver - Amazon
- Option: 3.5mm Jack Port - Amazon

Other Hardware

- M2 Screws Amazon and M3 Screws Amazon M3&Tripod Inserts Amazon
- Jumper Wires Amazon
- USB Cables (15-30cm)
    - Charge (USB-C to USB-C 90˚) - Amazon
    - Power (USB-A to USB-C 90˚) - Amazon
    - GPS (USB_A to Micro-USB) - Amazon



Allsky GO

Other Resources & Links

ChrisE | Printables.com – Printables Store

Allsky Go on GitHub - Custom Scrips for Allsky Go!

AllskyTeam · GitHub – Allsky Software Repository & Installation Guidelines



Allsky GO